

Four regression models. Only the first two are fitted by solving $(X^T X)b = X^T y$

simple linear
 $y = \beta_0 + \beta_1 x$

polynomial
 $y = \beta_0 + \beta_1 x + \beta_2 x^2$ (still least squares)

nonlinear in the parameters
 $y = \theta_1 e^{-\theta_2 x}$ (no closed form)

binary response: logistic
(Lecture 10, not least squares)

